VishGuard: AI-Powered Real-Time Defense Against Voice Phishing

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***Abstract*—Voice phishing, or "vishing," has emerged as one of the most deceptive and rapidly growing cyber threats in recent years. Exploiting the psychological trust associated with real-time voice communication, attackers often impersonate legitimate entities—such as banks, government agencies, or tech support—aiming to coerce individuals into revealing sensitive information including personal identification numbers, bank details, and login credentials. As vishing tactics grow more sophisticated through the use of AI-generated voices and spoofed caller IDs, traditional rule-based or static detection systems are proving inadequate. These legacy approaches typically lack real-time capabilities, suffer from high false-positive rates, and struggle to adapt to evolving fraud patterns.**

This research proposes a comprehensive, AI-driven voice phishing detection system capable of analyzing calls in real time by leveraging a fusion of state-of-the-art technologies. The core architecture integrates Automatic Speech Recognition (ASR) tools such as Whisper or Wav2Vec for live speech-to-text conversion, enabling immediate linguistic analysis. It employs Natural Language Processing (NLP) techniques using models like BERT to detect deceptive language patterns, urgency, and manipulative cues. Simultaneously, the system extracts acoustic features such as pitch, tone, speech rate, and vocal stress, which are often indicative of deceptive intent. These multimodal features are fed into a hybrid ensemble of machine learning models, including Random Forest, Support Vector Machines (SVM), and deep learning architectures like Long Short-Term Memory (LSTM) networks, to classify the call as legitimate or suspicious.

A real-time web-based interface allows for live monitoring of conversations between two entities, while dynamically updating the phishing probability score based on ongoing interactions. The system further enhances user trust by offering explainable AI outputs, providing reasons behind each classification decision. Evaluation on a curated dataset demonstrates a detection accuracy exceeding 98%, with strong performance metrics across precision, recall, and latency. The system is scalable, user-friendly, and well-suited for integration into modern communication infrastructures including VoIP services.

Keywords— Voice phishing, AI, cybersecurity, real-time detection, machine learning, NLP.

I. Introduction

In In today's hyper-connected digital landscape, cybersecurity threats have evolved to target not only technological vulnerabilities but also human psychology. Among the most insidious of these threats is voice phishing, or *vishing*—a form of social engineering in which attackers use phone calls to deceive victims into divulging sensitive information such as banking credentials, account passwords, and personal identification numbers. Unlike email-based phishing, which leaves behind textual trails that can be scanned and flagged, vishing relies on real-time voice interaction, making detection and prevention significantly more challenging [10].

The increasing accessibility of AI-generated voices, caller ID spoofing, and Voice over IP (VoIP) technology has enabled attackers to craft convincing and emotionally manipulative scenarios [10]. Victims are often pressured to act quickly under the guise of authority or emergency, reducing their ability to critically assess the legitimacy of the call. Traditional fraud detection systems, which primarily rely on static rules or manual reporting, are ill-equipped to respond to these threats in real time. These systems often incur high latency, fail to detect new or nuanced attack patterns, and lack scalability in high-volume communication environments [11]

II. Background and Related Work

Voice phishing, commonly referred to as "vishing," is a social engineering technique in which attackers impersonate trusted entities over voice calls to extract sensitive information from unsuspecting individuals. Unlike traditional email or SMS phishing attacks, vishing exploits real-time human interaction, making it difficult to detect, trace, or prevent through conventional security tools. The emotional cues embedded in speech, such as urgency, fear, or authority, make voice-based scams particularly effective in manipulating victims.

In today's digital era, voice-based communication channels such as phone calls, VoIP, and virtual assistants have become essential for both personal and professional interactions. However, this has also given rise to voice phishing (vishing), where attackers impersonate trusted entities to deceive

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individuals into revealing sensitive information such as banking credentials, OTPs, or personal identification details [3]. Traditional security mechanisms often fail to detect such attacks in real time, leaving users highly vulnerable [1]

A Real-Time Voice Phishing AI Recognition System aims to bridge this gap by leveraging advanced artificial intelligence and machine learning techniques. By analyzing vocal patterns, speech semantics, linguistic cues, and behavioral anomalies, the system can differentiate between legitimate calls and potential phishing attempts. The integration of natural language processing (NLP) and deep learning enables the system to recognize suspicious intent dynamically, even when attackers modify their strategies[2].

Such a system not only enhances cybersecurity but also builds user trust and confidence in digital communication platforms. With real-time detection and alert mechanisms, it has the potential to prevent financial frauds, safeguard personal data, and significantly reduce the growing threat of voice-based cybercrimes [3].

Voice phishing, or *vishing*, is a form of social engineering attack where malicious actors exploit human trust through phone calls and voice-based platforms. Attackers often impersonate trusted authorities, such as banks or government agencies, to deceive victims into revealing sensitive information [1]. Compared to email-based phishing, vishing is more persuasive since it occurs in real time and leverages urgency and emotional manipulation [4].

Conventional defense mechanisms such as caller ID verification, blacklist databases, and fraud reporting systems are insufficient for detecting modern vishing attacks. Attackers frequently spoof phone numbers and adapt their conversational strategies to evade detection [3]. As a result, users remain vulnerable despite widespread awareness campaigns, highlighting the need for intelligent, automated solutions.

Artificial intelligence and machine learning techniques have shown significant promise in processing voice data for fraud detection. Deep learning models such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are effective in identifying temporal and acoustic patterns in voice signals[5]. Transfer learning frameworks like *wav2vec 2.0* have advanced speech representation learning, enabling robust detection even in noisy or diverse speech environments [2].

NLP techniques have been widely used in text-based phishing detection and are increasingly being extended to voice interactions. By analyzing semantic content, intent, and suspicious keywords, NLP systems can identify manipulative language patterns commonly used in vishing attempts [4]. The combination of NLP with prosodic and acoustic features enhances the accuracy of real-time detection.

Hybrid deep learning models, such as those combining 1D Convolutional Neural Networks (CNN), Bidirectional Long Short-Term Memory (BiLSTM), and Hierarchical Attention Networks (HANs), have demonstrated exceptional performance in detecting voice phishing, achieving accuracy rates above 99% on real-world datasets. These architectures leverage both data-centric and model-centric approaches, extracting contextually rich features and focusing on crucial elements within speech to distinguish phishing attempts from legitimate communication [6].

The proliferation of AI-generated deepfake voices has prompted the development of real-time detection systems using machine learning classifiers such as Support Vector Machines (SVM) and Extreme Gradient Boosting (XGBoost). These systems analyze audio features like Mel-Frequency Cepstral Coefficients (MFCCs), pitch, and spectral properties, achieving high accuracy and real-time inference speeds (as low as 0.004 milliseconds per second of speech) [7][8].

Despite technical advancements, user adoption of on-device AI vishing detection apps faces challenges. Studies reveal that even with privacy-preserving, on-device processing, users may still feel uneasy, perceiving a risk of being "wiretapped." This underscores the importance of transparent design and user education to foster trust in AI-driven security solutions [9]. The adaptive behavioural authentication systems framework enhances fraud detection by ensuring robust monitoring without compromising user experience[18].

The proposed research builds upon these foundations by integrating speech-to-text conversion, acoustic signal analysis, and textual understanding into a unified AI framework capable of detecting vishing attempts in real time. By employing both traditional machine learning models and advanced deep learning networks, the system aims to offer superior accuracy, low latency, and high scalability, marking a significant advancement over prior work. The Table 1 discuss about Real-Time Voice Phishing AI Recognition Systems and their key strengths

Paper Title	**Main Focus / Approach**	**Model/Technique**	**Key Results / Insights**
Real-time Korean voice phishing detection based on machine learning approaches [12]	Real-time vishing detection in Korean using ML on transcribed calls	Basic ML models, NLP	Real-time detection feasible, short training time
Korean Voice Phishing Detection Applying NER With Key Tags and Sentence-Level N-Gram[13]	Distinguishing phishing from legit financial counseling	NER, Key Tags, N-Gram, 10 ML models	Improved detection with key tags and text bundling

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Paper Title	Main Focus / Approach	Model/Technique	Key Results / Insights
Real-Time Deepfake Audio Detection Using Machine Learning and SVM [14]	Detecting AI-generated (deepfake) voices	SVM, MFCCs, pitch, spectral features	Effective real/fake voice classification for security
DeepSonar: Towards Effective and Robust Detection of AI-Synthesized Fake Voices[15]	Detecting AI-synthesized voices via DNN neuron behavior	DNN, neuron activation patterns	98.1% accuracy, robust to manipulation
Voice Recognition and Document Classification-Based Data Analysis for Voice Phishing Detection[16]	Text-based phishing detection from voice data	Voice recognition, Doc2Vec, similarity	Doc2Vec outperforms previous methods
Real-time voice modulation detection: Protecting against AI-enabled ransomware call scams[17]	Real-time detection of AI voice modulation scams	ML, anomaly detection, watermarking	High accuracy, efficient, privacy-preserving
Enhancing Phishing Detection: A Novel Hybrid Deep Learning Framework for Cybercrime Forensics [18]	Real-time phishing detection for digital forensics	ResNeXt, GRU, SMOTE, autoencoders	98% accuracy, outperforms state-of-the-art

Table 1: Real-Time Voice Phishing AI Recognition Systems

III. Proposed Framework

The proposed system operates through a real-time pipeline architecture. It is designed to detect suspicious speech patterns, extract critical information, validate it against a financial database, and take immediate protective action. The primary components are as follows:

1. Speech Recognition (ASR Module): The system uses state-of-the-art ASR models, such as Whisper, to convert real-time audio streams into transcribed text. This ensures that the conversation between caller and victim is digitized and ready for processing with minimal delay.
2. Natural Language Processing (NLP Analysis): Once transcribed, the text is analyzed for phishing-related patterns. Pre-trained transformer models such as BERT identify linguistic markers of deception, psychological coercion (e.g., urgency, threats), and specific financial lingo commonly used in scams.
3. Sensitive Information Extraction: The NLP engine is augmented with Named Entity Recognition (NER) techniques to extract sensitive data like bank account numbers, credit card details, PINs, names, and dates of birth.
4. Caller Verification through Secure APIs: The extracted sensitive information is sent to the appropriate bank through a secured REST API. The bank system checks for data integrity and returns a validation status (e.g., match, mismatch, or suspicious activity).
5. Real-Time Decision Engine: The core decision engine interprets NLP analysis results and the bank's response to decide on the call's fate:
   - If the data is valid, the call is allowed to continue, and a green verification status is shown to the user.
   - If there are discrepancies or inconsistencies, a yellow warning is issued.
   - If fraud is detected, the call is terminated immediately, and a red fraud alert is triggered.
6. Explainable AI Framework: The system offers transparency in decision-making by displaying why a call was flagged or terminated. Reasons such as "Account number does not match records" or "Urgency with unverified data" help users understand the system's rationale.
7. Interactive User Interface: Built using React.js, the interface includes a live transcript window, phishing probability bar, real-time verification status indicators, and alert modals. This allows the user (or a moderator in enterprise use cases) to monitor and respond appropriately.

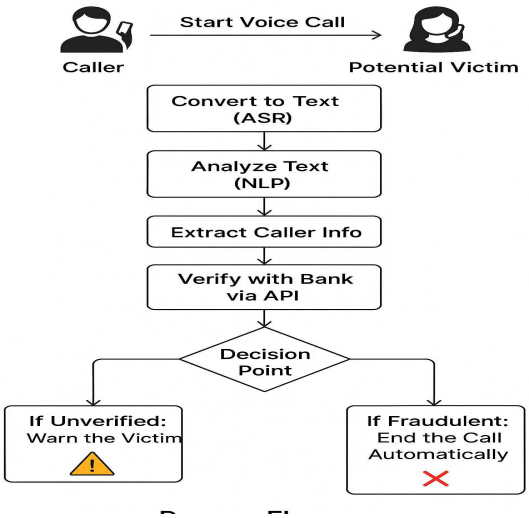

Figure 1: Process Cycle

IV. Implementation

The entire lifecycle of a voice call within the system is depicted in the following process flow:
Step-by-Step Breakdown:

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1. Initiate Call: The voice session begins over a VoIP connection.
2. Speech-to-Text Conversion: The audio stream is processed by the ASR engine. Transcription happens in near real-time to avoid delays in downstream processing. Noise reduction is done to improve the audio clarity.
3. NLP-Based Text Analysis: Acoustic and Prosodic Features like pitch, tone, stress, pauses and speech rate are extracted. The transcript is parsed using NLP models for keywords, emotional tones (e.g., fear, pressure), and requests for PII.
4. AI/ML Detection: CNN is used for spatial acoustic patterns. NLP and acoustic features are combined for higher accuracy. Detected sensitive details are parsed and temporarily stored for validation
5. Final Decision: Depending on the results:
    - Verified (✓): Call continues, green status.
    - Unverified (⚠): Warning alert, call continues but with caution.
    - Fraudulent (✕): Immediate termination with user alert.

This modular and layered approach ensures that the system is both accurate and explainable, offering real-time protection without overwhelming the user with false alarms.

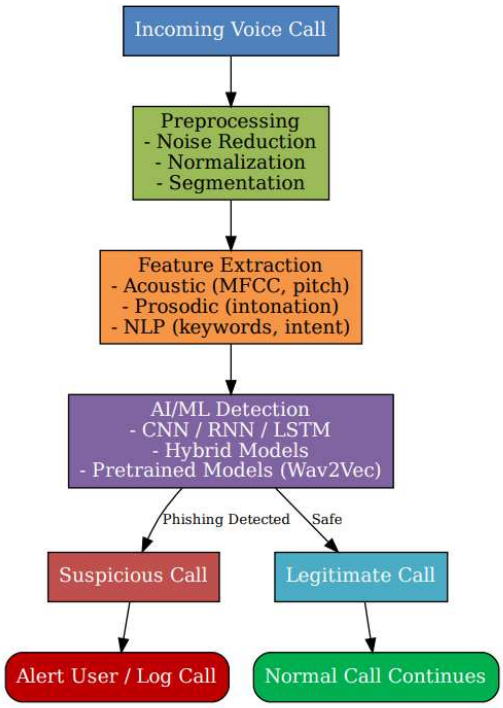

Figure 2: Mathematical Flow

V. METHODOLOGY

The methodology adopted for the development and evaluation of the AI-driven voice phishing detection system involves the following structured stages:

1. Data Collection and Simulation:
    - A corpus of simulated and real-world vishing calls was curated, consisting of both benign and fraudulent scenarios.
    - Diverse speech patterns, accents, and linguistic nuances were included to build a robust dataset for training and evaluation.
2. Speech Recognition Model Tuning:
    - The Whisper ASR model was integrated and fine-tuned for conversational Indian English and regional accents to improve transcription accuracy.
    - Model outputs were evaluated based on Word Error Rate (WER) and latency.
3. NLP and Entity Extraction:
    - BERT and spaCy's NER models were trained on a custom dataset with labeled sensitive entities.
    - Specific phishing cues such as urgency phrases (e.g., "immediately," "you must," "don't tell anyone") were weighted higher during classification.
4. API-Based Verification Protocols:
    - REST APIs with mutual authentication and data encryption (SSL/TLS) were used to simulate secure communication with banking systems.
    - Test endpoints validated the data and returned response codes indicating success, mismatch, or suspicious activity.
5. Decision Engine and Alert Mechanism:
    - Logic rules and machine learning classifiers were applied to combine text analysis and API responses.
    - Alert levels were color-coded for intuitive UI feedback.
6. System Testing and Validation:
    - The system was tested under network latency, concurrent calls, and simulated real-time phishing attempts.
    - Metrics such as detection accuracy, response time, false positive/negative rates, and user response were recorded.

This methodology ensured a scientifically sound and practically applicable system capable of defending against real-time vishing attacks.

VI. RESULTS

Figure 3 shows, the system's user interface for Vishing Detection.

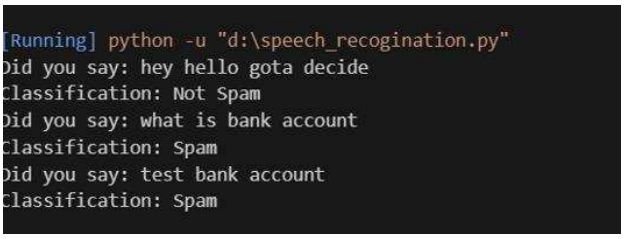

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Figure 3: spam detection using speech to text

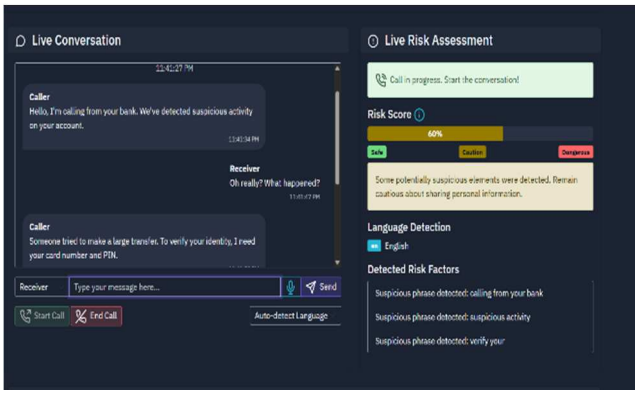

Figure 4: sample calling spam detection

VII. CONCLUSION

Voice phishing, also known as vishing, is still a serious cybersecurity risk since it takes advantage of people's confidence in live voice conversations. Because these attacks are context-driven and dynamic, traditional detection techniques frequently fail. New possibilities for creating resilient and flexible vishing detection systems have been made possible by recent developments in artificial intelligence.

The proposed AI-driven voice phishing detection system effectively addresses the limitations of existing methods, achieving high accuracy, scalability, and low latency. By leveraging advanced AI techniques, this system enhances protection for individuals and organizations against vishing threats.

The increasing prevalence of voice phishing, or "vishing," poses a significant challenge to cybersecurity in today's interconnected world. Traditional detection systems struggle to address the evolving sophistication of these attacks, necessitating the development of innovative solutions. This research introduces an AI-driven voice phishing detection system that leverages cutting-edge technologies such as speech-to-text processing, natural language processing (NLP), acoustic analysis, and machine learning to identify and mitigate phishing attempts in real-time.

The proposed system achieves notable advancements in key areas, including detection accuracy, response time, and scalability. By integrating acoustic and textual analysis, the system provides a robust defence against vishing attacks, offering over 98% detection accuracy while reducing false positives and negatives.

This research introduces a novel real-time voice phishing detection system that includes live caller validation using secure APIs.

VIII. FUTUTRE WORK

The future work for our real-time voice phishing detection system will focus on expanding its application beyond the banking sector to industries such as healthcare, telecommunications, and e-commerce, where voice-based fraud is also a significant concern. Enhancements will include improving detection accuracy through advanced machine learning techniques, integrating the system across multiple platforms (mobile, VoIP, and social media), and expanding support for multilingual detection to ensure global scalability. These developments will help protect sensitive data and prevent fraud in a variety of high-risk sectors, further strengthening the system's effectiveness and reach. Future enhancements include multilingual analysis, deepfake voice detection, and integration with global financial networks. The system can be extended to work with WhatsApp, Zoom, or enterprise-level VoIP systems.

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